

Question No:4

DS

Session

Sorting

Question Information

Level 1

Challenge 14

Question description

APPU needed a laptop, so he went to a neighbouring garage sale.

At a sale, there were n laptops.

The cost of a laptop with index i is a_i rupees.

Some laptops have a negative price, meaning their owners are willing to pay APPU if he buys their broken laptop. APPU is free to purchase whichever laptop he desires.

Despite his strength, he can only handle m Laptops at a time, and he has no desire to return to the auction.

Please, help APPU find out the maximum sum of money that he can earn.

Constraints:

$$1 \leq T \leq 10$$

$$1 \leq n, m \leq 100$$

$$-1000 \leq a_i \leq 1000$$

Input Format:

First line of the input contains T denoting the number of test cases. Each test case has 2 lines :

first line has two spaced integers n m .

second line has n integers $[a_0 \dots a_{i-1} \dots a_{n-1}]$.

Output Format:

The maximum sum of money that LALU can earn, given that he can carry at most m Laptops.

Logical Test Cases

Test Case 1

INPUT (STDIN)

1
5 3
-6 0 35 -2 4

EXPECTED OUTPUT

8

Test Case 2

INPUT (STDIN)

2
6 2
-4 5 0 -1 2 1
7 4
6 1 0 -2 2 -4 3

EXPECTED OUTPUT

5
6

Mandatory Test Cases

Test Case 1

KEYWORD

```
int MEGA_SALE(int arr[],int  
no,int k)
```

Test Case 2

KEYWORD

```
void bubble_sort(int  
arr[],int no)
```

Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

4

Test Case 2

TOKEN COUNT

320

Test Case 3

NLOC

55

Code:

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  void bubble_sort(int arr[], int n)
4  {
5      int i, j, temp;
6
7      for (i = 0; i < n - 1; i++)
8      {
9          for (j = i + 1; j < n; j++)
10         {
11             if (arr[i] > arr[j])
12             {
13                 temp = arr[i];
14                 arr[i] = arr[j];
15                 arr[j] = temp;
16             }
17         }
18     }
19 }
20
21 int main()
22 {
23     int T;
24     scanf("%d", &T);
25
26     while (T--)
27     {
28         int n, m;
```

```
29     scanf("%d %d", &n, &m);
30
31     int a[n];
32
33     for (int i = 0; i < n; i++)
34     {
35         scanf("%d", &a[i]);
36     }
37
38     bubble_sort(a, n);
39
40     int sum = 0;
41     int count = 0;
42
43     for (int i = 0; i < n && count < m; i++)
44     {
45         if (a[i] < 0)
46         {
47             sum += -a[i];
48             count++;
49         }
50         else
51         {
52             break;
53         }
54     }
55     printf("%d\n", sum);
56 }
57 return 0;
58 }
```

Output:

Test Case-1:

```
1
5 3
-6 0 35 -2 4
8
```

Test Case-2:

```
2
6 2
-4 5 0 -1 2 1
5
7 4
6 1 0 -2 2 -4 3
6
PS C:\Users\kesha\OneDrive\Documents\3sem\C>
```

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